

Chapter 6: Symmetries

Quantum Mechanics

1 Introduction to Symmetry

Definition 1.1 (Symmetry). [def:symmetry] A *symmetry* of a system is some transformation, such that the system is invariant before and after the transformation.

Definition 1.2 (Symmetry of Quantum System). [def:symmetry-qm] A *symmetry* of a quantum system is a transformation that leaves the Hamiltonian invariant.

Definition 1.3 (Types of Symmetry). [def:symmetry-types] There are several types of symmetries in quantum mechanics, including:

- *Translation symmetry*: The system is invariant under some spatial translations.
- *Rotation symmetry*: The system is invariant under some rotations.
- *Reflection symmetry*: The system is invariant under reflections on some axis/planes.
- *Inversion symmetry*: The system is invariant under inversion of all coordinates.
- *Time-reversal symmetry*: The system is invariant under the reversal of time.

Theorem 1.1 (Symmetry Condition on the Hamiltonian). [thm:symmetry-condition-hamiltonian] An operator $\hat{U}$ realizes a definition 1.2 of a system with Hamiltonian $\hat{H}$ if and only if $\hat{H}$ is invariant under definition conjugation action:

$$\hat{U}^\dagger \hat{H} \hat{U} = \hat{H}, \quad \text{equivalently (for linear } \hat{U} \text{)} \quad [\hat{U}, \hat{H}] = 0.$$

For antiunitary $\hat{U}$ (e.g. time reversal), the second form must be replaced by the appropriate anti-linear condition.

2 Translation Operator

Definition 2.1 (Translation Operator). [def:translation-operator] Intuitively, a *translation operator* $\hat{T}(\vec{r}_0)$ (translates a state by a vector $\vec{r}_0$) acts on a state $|\psi\rangle$ should let $|\psi\rangle$ be translated by a vector $\vec{r}$, i.e.

$$\hat{T}(\vec{r}_0) |\psi\rangle = |\psi(\vec{r} - \vec{r}_0)\rangle$$

Definition 2.2 (Translation Operator (1D)). [def:translation-operator-1d] In one dimension, the translation operator $\hat{T}(a)$ shifts a state by a real distance $a \in \mathbb{R}$. Its action on a position-space wavefunction is

$$\hat{T}(a) \psi(x) = \psi(x - a).$$

Theorem 2.1 (Exponential Form of the Translation Operator). [thm:translation-exponential] The one-dimensional translation operator can be written as:

$$\hat{T}(a) = e^{-ia\hat{p}/\hbar}$$

Proof. Expand the exponential in a Taylor series at 0

$$\hat{T}(a) = e^{-ia\hat{p}/\hbar} = \sum_{n=0}^{\infty} \frac{(-ia/\hbar)^n}{n!} \cdot \hat{p}^n = \sum_{n=0}^{\infty} \frac{(-ia\hat{p})^n}{n!\hbar^n}$$

According to the definition of the momentum operator, $\hat{p} = -i\hbar \frac{d}{dx}$, so we can substitute this into the series expansion:

$$\hat{T}(a) = \sum_{n=0}^{\infty} \frac{(-ia(-i\hbar \frac{d}{dx}))^n}{n!\hbar^n} = \sum_{n=0}^{\infty} \frac{(-a(\frac{d}{dx}))^n}{n!}$$

So,

$$\hat{T}(a)\psi(x) = \sum_{n=0}^{\infty} \frac{(-a(\frac{d}{dx}))^n}{n!} \psi(x) = \sum_{n=0}^{\infty} \frac{(-a)^n}{n!} \psi^{(n)}(x)$$

Then, we do a Taylor expansion of $\psi(x)$ centered at $x + a$:

$$\psi(x) = \sum_{n=0}^{\infty} \frac{\psi^{(n)}(x+a)}{n!} (x - (x+a))^n = \sum_{n=0}^{\infty} \frac{\psi^{(n)}(x+a)}{n!} (-a)^n = \sum_{n=0}^{\infty} \frac{(-a)^n}{n!} \psi^{(n)}(x+a)$$

Substituting $x \rightarrow x - a$ in the previous identity,

$$\psi(x-a) = \sum_{n=0}^{\infty} \frac{(-a)^n}{n!} \psi^{(n)}((x-a)+a) = \sum_{n=0}^{\infty} \frac{(-a)^n}{n!} \psi^{(n)}(x).$$

Which means,

$$\hat{T}(a)\psi(x) = \psi(x-a)$$

□

Theorem 2.2 ($\hat{T}(a)$ is a Unitary Operator). [thm:translation-unitary] *The translation operator $\hat{T}(a)$ is a unitary operator, i.e. $\hat{T}^\dagger(a) = \hat{T}^{-1}(a) = \hat{T}(-a)$.*

Remark 2.1 (Intuition about). [rem:intuition-about]

Proof. **TODO:** Proof to be written after clarifying preferred approach with the lecturer — direct from the action definition (definition 2.2), or via the exponential form (theorem 2.1). Email sent September 22, 2026. □